Computer Graphics

Week 3 : *Bressenhem (Circle)*

A circle is a collection of points that have a distance from the same center point for all points.

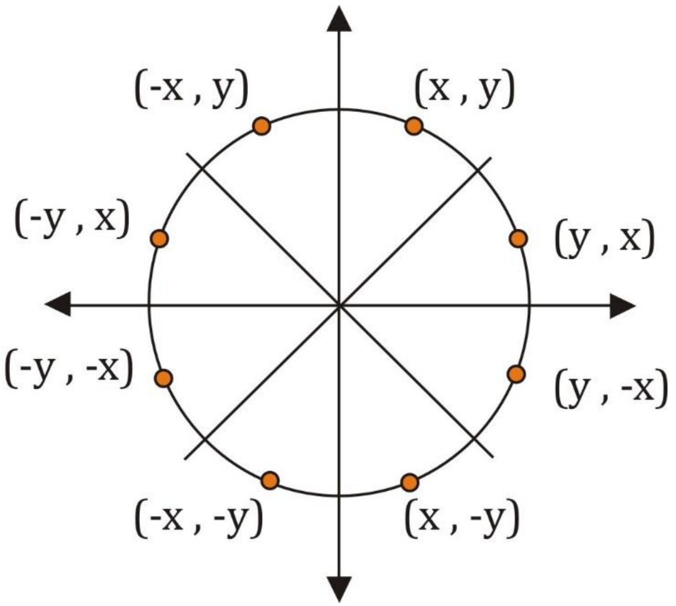
In general, circles are used as components of an image. The procedure for displaying circles and ellipses is made with the basic equation of the circle $x^2 + y^2 = r^2$.

Circles can be made by drawing a quarter circle, because other parts can be made as symmetrical parts.

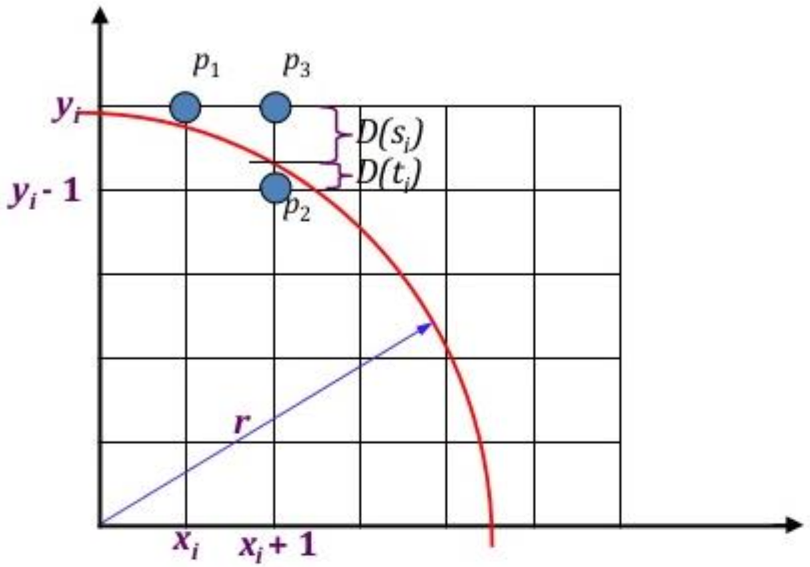
Symmetrical Eight Points

The process of making a circle can be done by determining one starting point. If the starting point is a circle (x, y), then there are three other positions, so that eight points can be obtained.

Thus, it is only necessary to calculate the segment 45^0 in determining the complete circle.



The algorithm used forms all points based on the center point with the addition of all the paths around the circle.



F_{circle} (x,y) ; < 0, if (x,y) inside the circle line

F_{circle} (x,y) ; = 0, if (x,y) on the circle line

F_{circle} (x,y) ; > 0, if (x,y) outside the circle line

$P_1 = M(x_k, y_k)$

$P_2 = M(x_k + 1, y_k - 1)$

$P_3 = M(x_k + 1, y_k)$

After point P_1, do we choose P_2 or P_3 ?

$D(s_i)$ = distance of P_3 from circle → $D(s_i) = (x_k + 1)^2 + y_k^2 - r^2$

$D(t_i)$ = distance of P_2 from circle → $D(t_i) = (x_k + 1)^2 + (y_k - 1)^2 - r^2$

Decision Parameter P_k $= D(s_i) + D(t_i)$

$= (x_k + 1)^2 + y_k{}^2 - r^2 + (x_k + 1)^2 + (y_k - 1)^2 - r^2$

$= 2(x_k + 1)^2 + y_k{}^2 + (y_k - 1)^2 - 2r^2$......... (1)

So if $P_k < 0$ the circle is closer to P_3 (above point)

$(x_k + 1, y_k) = (x_k + 1, y_k)$

So if $P_k \geq 0$ the circle is closer to P_2 (below point)

$(x_k + 1, y_k) = (x_k + 1, y_k - 1)$

Next decision parameter P_{k+1} :

$P_{k+1} = 2(x_{k+1} + 1)^2 + y_{k+1}{}^2 + (y_{k+1} - 1)^2 - 2r^2$......... (2)

Difference between equation (2) – (1)

$$P_{k+1} - P_k = 2(x_{k+1} + 1)^2 + y_{k+1}^2 + (y_{k+1} - 1)^2 - \cancel{2r^2}$$
$$- 2(x_k + 1)^2 - y_k^2 - (y_k - 1)^2 + \cancel{2r^2}$$

where $x_{k+1} = x_k + 1$

$$P_{k+1} - P_k = 2(x_k + 1 + 1)^2 + y_{k+1}^2 + (y_{k+1} - 1)^2$$
$$- 2(x_k + 1)^2 - y_k^2 - (y_k - 1)^2)$$
$$= 2(x_k + 2)^2 + y_{k+1}^2 + (y_{k+1} - 1)^2$$
$$- 2(x_k + 1)^2 + y_k^2 + (y_k - 1)^2$$
$$= 2(x_k^2 + 4x_k + 4) + y_{k+1}^2 + y_{k+1}^2 - 2y_{k+1} + \cancel{1}$$
$$- 2(x_k^2 + 2x_k + 1) - y_k^2 - y_k^2 + 2y_k - \cancel{1}$$
$$= \cancel{2x_k^2} + 8x_k + 8 + 2y_{k+1}^2 - 2y_{k+1}$$
$$\cancel{-2x_k^2} - 4x_k - 2 - 2y_k^2 + 2y_k$$
$$= 4x_k + 6 + 2y_{k+1}^2 - 2y_{k+1} - 2y_k^2 + 2y_k$$

$$P_{k+1} = P_k + 4x_k + 2(y_{k+1}^2 - y_k^2) - 2(y_{k+1} - y_k) + 6$$

If $P_k < 0$ (P_3 is select) then $y_{k+1} = y_k$

$$P_{k+1} = P_k + 4x_k + 2(\cancel{y_k^2 - y_k^2}) - 2(\cancel{y_k - y_k}) + 6$$

$$= P_k + 4x_k + 6$$

If $P_k \geq 0$ (P_2 is select) then $y_{k+1} = y_k - 1$

$$P_{k+1} = P_k + 4x_k + 2((y_k - 1)^2 - y_k^2) - 2(y_k - 1 - y_k) + 6$$

$$= P_k + 4x_k + 2(\cancel{y_k^2} - 2y_k + 1 \cancel{- y_k^2}) - 2(\cancel{y_k} - 1 \cancel{- y_k}) + 6$$

$$= P_k + 4x_k - 4y_k + 10$$

Initial decision Parameter P_0

$D(s_i)$ = distance of P_3 from circle → $D(s_i) = (x_k + 1)^2 + y_k^2 - r^2$

$D(t_i)$ = distance of P_2 from circle → $D(t_i) = (x_k + 1)^2 + (y_k - 1)^2 - r^2$

$x_k = 1$

$y_k = r$

$$
\begin{aligned}
P_0 &= D(s_i) + D(t_i) \\
&= [1^2 + \cancel{r^2} \cancel{- r^2}] + [1^2 + (r - 1)^2 - r^2] \\
&= 1 + 1 + \cancel{r^2} - 2r + 1 \cancel{- r^2} \\
&= 3 - 2r
\end{aligned}
$$

Circle Formation Bressenhem Steps:

1. Determine the radius *r* with the center point of the circle (x_c, y_c); $y_c = r \rightarrow (0,r)$
2. Calculate the parameter $p_0 = 3 - 2r$
3. While $(x < y)$, Determine the initial value $k = 0$ for each x_k position:
   a) If $p_k < 0$ then the next point is $(x_k + 1, y_k)$ and $p_{k+1} = p_k + 4x_k + 6$
   b) If $p_k > 0$ then the next point is $(x_k + 1, y_k - 1)$ and $p_{k+1} = p_k + 4(x_k – y_k) + 10$
   Where $2x_{k+1} = 2x_k + 2$ and $2y_{k+1} = 2y_k - 2$
4. Determine the symmetrical point on the other seven octane.
5. Move each pixel position (x, y) on the circular line of the circle with the center point (x_c, y_c) and specify the coordinate value:
   $x = x + x_c$ and $y = y + y_c$
6. Repeat steps 3 -5, up to $x \geq y$

Example

To illustrate the Bressenham algorithm in the formation of a circle with a center point (0,0) and a radius is 10, the calculation is based on the first quadrant where x = 0 to x = y

Iteration 1

$k = 0$

$p_0 = 3 - 2r \rightarrow p_0 = 3 - 20 = -17$

$p_0 < 0$

$(x_k + 1, y_k) \rightarrow (x_0 + 1, y_0) \rightarrow (1, 10)$

$p_{k+1} = p_k + 4x_k + 6 \rightarrow p_1 = -17 + 4.0 + 6 \rightarrow p_1 = -11$

$2x_{k+1} = 2x_k + 2$ and $2y_{k+1} = 2y_k - 2$

$2.1 = 2.0 + 2$

Iteration 2

$k = 1$

$p_1 = -11$

$p_1 < 0$

$(x_k + 1, y_k) \rightarrow (x_1 + 1, y_1) \rightarrow (2, 10)$

$p_{k+1} = p_k + 4x_k + 6 \rightarrow p_1 = -11 + 4.1 + 6 \rightarrow p_2 = -1$

$2x_{k+1} = 2x_k + 2$ and $2y_{k+1} = 2y_k - 2$

$2.2 = 2.1 + 2$

Iteration 3

$k = 2$

$p_2 = -1$

$p_2 < 0$

$(x_k + 1, y_k) \rightarrow (x_2 + 1, y_2) \rightarrow (3, 10)$

$p_{k+1} = p_k + 4x_k + 6 \rightarrow p_1 = -1 + 4.2 + 6 \rightarrow p_3 = 13$

$2x_{k+1} = 2x_k + 2$ and $2y_{k+1} = 2y_k - 2$

$2.3 = 2.2 + 2$

Iteration 4

$k = 3$

$p_3 = 13$

$p_3 > 0$

$(x_k + 1, y_k - 1) \rightarrow (x_3 + 1, y_3 - 1) \rightarrow (4, 9)$

$p_{k+1} = p_k + 4(x_k - y_k) + 10 \rightarrow p_4 = 13 + 4(3 - 10) + 10 \rightarrow p_4 = -5$

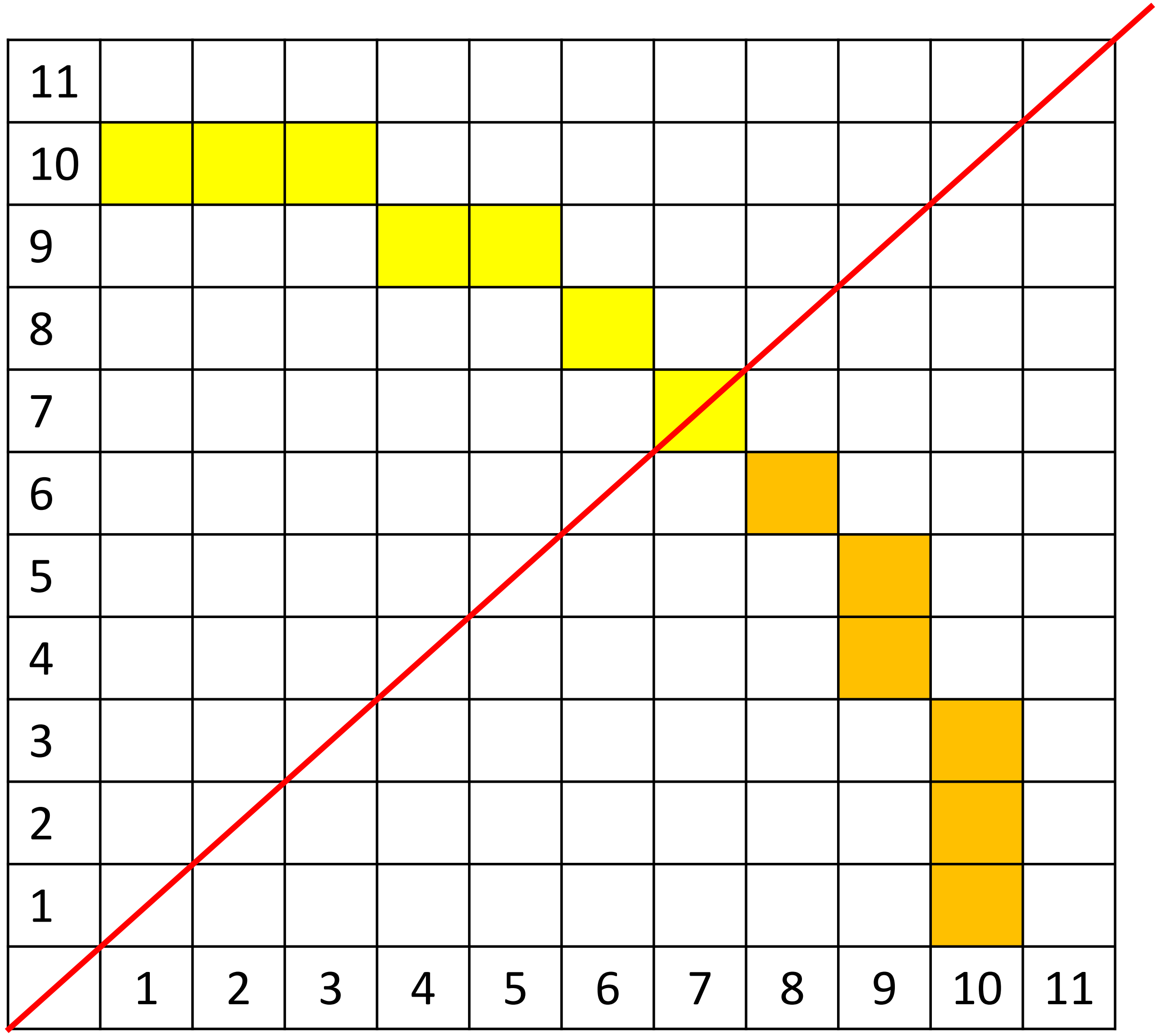
$2x_{k+1} = 2x_k + 2$ and $2y_{k+1} = 2y_k - 2$

$2.9 = 2.10 - 2$

Iteration 4

and so on. . .

k	Pk	(x_{k+1}, y_{k+1})
0	-17	(1,10)
1	-11	(2,10)
2	-1	(3,10)
3	13	(4,9)
4	-5	(5,9)
5	17	(6,8)
6	11	(7,7)



EXERCISE

Create a circle with $r = 9$ towards the center point $(0,0)$

THANKYOU